



SESSION 15

Linear Algebra Roadmap

The Foundations of Vector Spaces · Matrix Computations ·
Applications in Deep Learning

Why Linear Algebra in Machine Learning?

Vector Spaces & Basic Definitions

Matrix Calculations & Representations

Systems of Linear Equations

Eigendecomposition & Principal Components

Dimensionality Reduction & SVD

Deep Learning Applications (Tensors & Backprop)

Learning Pathway & Recommended Resources

Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Instructor: Nitish Singh (CampusX)



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01 • Why Linear Algebra in Machine Learning?

Linear Algebra is the language of modern computing. Machine learning datasets are represented as matrices, and models process features using vector spaces and tensor transformations.

02 · Vector Spaces & Basic Definitions

A vector space is a collection of objects (vectors) that can be added together and multiplied by scalars. Vectors represent coordinates in multi-dimensional space (e.g., individual records in a dataset).

03 • Matrix Calculations & Representations

Matrices are 2D arrays storing dataset features. Matrix-vector multiplication represents linear transformations, mapping features from one space (input layer) to another (hidden layers).

04 · Systems of Linear Equations

Many machine learning tasks (like solving ordinary least squares in linear regression) are mathematically formulated as solving systems of equations: $A\mathbf{x} = \mathbf{b}$.

05 • Eigendecomposition & Principal Components

Finding eigenvalues and eigenvectors reveals the direction of maximum variance in data. Eigendecomposition is the mathematical engine behind PCA (Principal Component Analysis).

06 • Dimensionality Reduction & SVD

Singular Value Decomposition (SVD) decomposes matrices into component matrices, facilitating data compression, noise reduction, and recommender system models.

07 · Deep Learning Applications (Tensors & Backprop)

Tensors generalise matrices to higher dimensions. Deep neural networks consist of matrix multiplications followed by non-linear activations. Backpropagation computes gradients using transpose matrices and chain rules.

08 • Learning Pathway & Recommended Resources

To master Linear Algebra for ML:

1. Vectors and vector operations (dot products, norms).
2. Matrices and linear transformations.
3. Solving $Ax = b$ systems.
4. Eigendecomposition and SVD.

09 · Quick Reference Card



Linear Algebra Cheat Sheet

- **Scalar:** Single number (0 -D).
- **Vector:** 1 -D array of numbers.
- **Matrix:** 2 -D array of numbers.
- **Tensor:** n -dimensional array of numbers.